

Week 8 Check-in

Current progress:

We have a working ARIMA model and LSTM model that takes as input 7 days of household energy consumption and predicts the next day's consumption. Both models currently use a randomly selected subset of households as the sample and try to make predictions for an unseen household. The MSE of the prediction is on the order of $e-02$.

We also have a clustering algorithm that finds 10 representatives for the entire set of households. The clustering algorithm first reduces the dimension of the data and then runs the k-means clustering algorithm.

Next steps:

1. Adjust the ARIMA model so that both the LSTM and ARIMA models compute MSE of the same subsets. Align MSE calculations for both models.
2. Find clusters of size 1, 5, 10, 15, and 20 so that we could compare if the number of clusters affects the result of the ARIMA and LSTM models.